

CS156 Project HOMIES: Machine Learning on a Home Task Activity Dataset

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Keywords: Human Activity Recognition (HAR), Machine Learning (ML), Feature Extraction, sound, accelerometer.

Abstract This project centers on a dataset focused on at-home tasks ranging from passively watching TV for entertainment to actively mopping to clean the floor. Accelerometer data was captured by a wearable device wristband, and environmental sound was captured by a smart-phone. Our project’s feature extraction was driven by hand-engineered features such as magnitude mean, magnitude root-mean-squared, and x root-mean-squared. We implemented classical ML models including XGBoost, Random Forest, and Naive Bayes and evaluated performance using both standard 80/20 train-test splits and Leave-One-Subject-Out (LOSO) validation. Our final tuned XGBoost model achieved 81.8% accuracy and 81.3% macro F1 under a standard 80/20 split, but performance dropped to 62.5% accuracy and 60.9% macro F1 under LOSO evaluation. This gap shows that random train-test splitting can overestimate real-world wearable HAR performance because windows from the same users may appear in both training and testing. Our window-size analysis showed that very short windows were less stable, while longer windows generally improved classification performance. The selected 3-second window was a reasonable trade-off between accuracy, stability, and number of training samples. Activities such as sweeping and mopping were often confused because they share repeated arm movements, while eating chips was harder to classify because a short window may capture either a rest period or a quick hand-to-mouth motion.

Overall, the results show that accelerometer-based HAR can recognize several home activities, but cross-user generalization remains challenging with only three participants and wrist accelerometer data alone. Future work could improve performance by adding audio features, increasing the number of participants, testing deep sequence models such as CNNs or LSTMs, and using longer temporal context to better capture activities with repeated or irregular motion.

1 Introduction

Human Activity Recognition (HAR) comprises of detecting and recognizing different types of human activity in events, behaviors, group actions, interactions, actions, and gestures. Gestures are the basic unit of activity such as a person raising their hand or sticking out their foot. Actions are single person activities based on multiple gestures such as stretching, typing, and punching. Interactions involve two or more people and/or objects, for example people shaking hands or 2 people writing on a paper with pencil. Group actions involve groups of people acting collectively

such as a group jumping and flipping, or a group listening to a speaker. Behaviors are activities that are based on a person's social, psychological, and emotional state such as nail biting from nervousness or jumping for joy. Events operate at a higher level and can "demonstrate a person's goal or social role and describe interpersonal social events." There can be more meaning to punching like shadow-boxing to train, writing to win in Tic Tac Toe, the group jumping and flipping performing a gymnastics team, and a class listening to the professor. These categories build on each other, where an event is based on the behaviors of the gymnastics team whose group action is made of interactions between team members and individual movements which are actions made of gestures. [1]

There are further classifications between activities concerning their level of energy use and variability. Energy use refers to the total amount of energy that was expended, it increases as the power needed for the activity increases. Variability indicates how much the movement changes over time (speed, direction, and force). As variability increases, repetition decreases (though can still function to accomplish the same goal of the activity). We define active tasks as tasks that have both a higher level of energy use and higher variability (e.g. washing hands and sweeping) in contrast to passive tasks which have lower energy use and are more repetitive (e.g. watching tv and typing on a keyboard). The fields that contribute and intersect in HAR are computer science (CS), computer vision (CV), data collection, signal processing (SP), socio-behavior analysis, machine Learning (ML), and more recently deep learning (DL). There are various methods of data collection from environmental devices such as video cameras to wearable devices such as smartwatches, wristbands, and smartphones. Today wearable devices and especially smartphones are ubiquitous, with most households in the U.S.A. owning one or more. These devices can come equipped with various sensors such as accelerometers, gyroscopes, barometers, pedometers, microphones, GPS, and heart rate monitors. The availability and ease of use that comes with wearable devices allows for vast collections of location independent, target-specific unlabeled data .

HAR feature extraction can be hand-engineered by experts using statistics such as area under curve (AUC), Mel-Frequency Cepstral Coefficients (MFCC), and F1-score. Comparatively expert-driven features can take more time to create than auto-generated features of DL. While hand-engineered features can be more useful for problems with sparse data and for being interpretable they can also be less accurate as well as tied to the specific application and not generalizable due in part to reliance on labeled data. DL features could require less human effort and time while capturing higher order data patterns.[2]

The applications of HAR are broad intertwined with study of Human-Computer Interaction. Major focuses include lifestyle/fitness monitoring:heartrate, steps, and nonstandard deviation as well as areas of improvement in dance/sport, surveillance: human behavior analysis, healthcare: disease detection and diagnosis. The variety of the types of data can be aggregated but they can also be used in more significant methods of data fusion. Multi-stacking, using base models as learners and a meta-model have been shown to have more accurate predictions using this dataset. [3]

This dataset focuses on at-home tasks ranging from passively watching TV for entertainment to actively mopping to clean the floor. Accelerometer data was captured by a wearable device wristband, and environmental sound was captured by a smartphone. Our project's feature extraction was driven by hand-engineered features such as magnitude mean, magnitude root-mean-squared, and x root-mean-squared. We implemented classical ML models including XGBoost, Random Forest, and Naive Bayes and evaluated performance using both standard 80/20 train-test splits and Leave-One-Subject-Out (LOSO) validation. Our final tuned XGBoost model achieved 81.8% accuracy and 81.3% macro F1 under a standard 80/20 split, but performance

dropped to 62.5% accuracy and 60.9% macro F1 under LOSO evaluation. This gap shows that random train-test splitting can overestimate real-world wearable HAR performance because windows from the same users may appear in both training and testing.

2 Literature Review

The paper HTAD: A Home-Tasks Activities Dataset with Wrist-accelerometer and Audio Features by Enrique Garcia-Ceja, Vajira Thambawita, Steven Hicks, Debesh Jha, Petter Jakobsen, Hugo L. Hammer, Pl Halvorsen, and Michael Riegler provides a novel dataset that collects both wrist-accelerometer and audio Features. Though it has comparatively fewer participants (3) and volume of data (23.1 minutes), HTAD produces results that contribute towards the fields of wearable and real time data collection, sensor data fusion, and home-task activity recognition. 7 activities such as watching TV, washing hands, eating chips, typing on a keyboard, brushing teeth, sweeping the floor, and mopping are performed by 25-30 year old participants (2 men and 1 woman). The authors calculate hand-engineered statistical accelerometer and audio (MFCC) features, perform baseline ML on the features separately and combined, and find that combining the data produces more accurate predictions. [4]

The paper "Multi-view stacking for activity recognition with sound and accelerometer data" by Enrique Garcia-Ceja, Carlos E. Galvn-Tejada, and Ramon Brena is actually a precursor to HTAD [4]. Using the same sound and accelerometer this project focuses on, Garcia-Ceja, Galvn-Tejada, and Brena apply concepts from multi-view learning and stacked generalization. Multi-view learning in this case involves taking advantage of the fact that different sensor types are complementary where different sensors each have different information that another sensor may not know. These sensors share that information with each other and learn from each other's predictions. Stacked generalization is the way this paper focused on combining the different views of audio and acceleration in contrast to mere aggregation of the views. In stacked generalization, each view is trained on with a base ML model (in this case, Random Forest for HAR) and their outputs are vectors that are concatenated with a vector that has the correct labels forming a matrix (new data to train on). A meta learner (still Random Forest), trains on that data and its predictions are used to evaluate performance. Garcia-Ceja, Galvn-Tejada, and Brena found that using this technique for this dataset as well as other datasets that utilized multi-modal sensors allowed for predictions to be greater in accuracy, specificity, and recall. [3]

The paper XGBoost-Based Human Activity Recognition Algorithm using Wearable Smart Devices by Runchan Ge proposes using XGBoost to classify human activities from wearable sensor data such as accelerometers and gyroscopes. The main idea is to improve accuracy while keeping the model efficient enough for wearable devices. The method includes preprocessing raw sensor signals, removing noise, extracting statistical features like mean and variance, and applying PCA for dimensionality reduction. These features are then fed into an XGBoost classifier, which uses gradient-boosted decision trees for activity recognition. The approach is typically evaluated on standard datasets like the UCI Human Activity Recognition dataset or similar smartphone sensor datasets. Results show that XGBoost achieves high accuracy and often outperforms traditional machine learning models such as SVMs and decision trees, while being faster and less resource-intensive than deep learning methods. However, limitations include reliance on handcrafted features, reduced generalization to new users or devices, and limited ability to distinguish similar activities. Overall, the paper shows that XGBoost is an effective and efficient solution for wearable-based human activity recognition. [5]

The paper Deep-Learning-Based Human Activity Recognition Using Wearable Sensors by A. Nouriani, R.A. McGovern, and R. Rajamani explores the use of deep learning to classify human activities from wearable sensor data, such as accelerometers and gyroscopes. The main idea is to

replace manual feature engineering with automatic feature learning from raw or lightly processed time-series data. The approach typically involves segmenting and normalizing sensor signals, then applying models such as Convolutional Neural Networks (CNNs) and sometimes LSTMs or RNNs to capture spatial and temporal patterns. A softmax layer is used for final activity classification, including actions like walking, sitting, standing, and climbing stairs. The model is evaluated on standard datasets such as the UCI Human Activity Recognition dataset and similar smartphone-based sensor datasets. Results show that deep learning models outperform traditional machine learning methods such as SVMs and decision trees in terms of accuracy and robustness, especially for complex activity patterns. However, the study also notes limitations, including high computational cost, reliance on large labeled datasets, limited interpretability, and reduced generalization to new users or devices. Overall, the paper shows that deep learning significantly improves HAR performance while introducing trade-offs in efficiency and interpretability. [6]

The paper Window Size Impact in Human Activity Recognition by Oresti Baños, Juan-Manuel Gálvez, Miguel Damas, Hector Pomares, and Ignacio Rojas investigates a question the HAR literature had largely left unanswered: how to choose the window size when segmenting accelerometer signals with the sliding-window approach. The main idea is that prior work used window sizes ranging from 0.25 s to over 10 s with no principled justification, and that this single choice propagates through the entire recognition pipeline and should be validated empirically. The method sweeps non-overlapping window sizes from 0.25 s to 7 s and evaluates recognition accuracy with several classifiers including decision trees, kNN, naive Bayes, and nearest-centroid, holding feature and classifier choices constant so the effect of window size can be isolated. The approach is evaluated on a 33-activity benchmark collected from subjects wearing multiple body-worn accelerometers. Results show that a 2.5 s window provides the best trade-off between accuracy and detection latency, with longer windows adding little accuracy while reducing both training-sample count and responsiveness. However, limitations include the fitness-oriented benchmark (jumping jacks, squats), which does not transfer cleanly to low-motion daily-living tasks, and the use of non-overlapping windows that leaves the interaction with overlap unstudied. Overall, the paper shows that window size should be treated as a hyperparameter swept empirically rather than fixed by convention, which directly motivates the window-size sweep used in our own windowing-strategy analysis. [7]

The paper A Public Domain Dataset for Human Activity Recognition Using Smartphones by Davide Anguita, Alessandro Ghio, Luca Oneto, Xavier Parra, and Jorge L. Reyes-Ortiz introduces the UCI HAR dataset and formalizes a now-standard preprocessing convention for accelerometer-based HAR: separating the raw signal into a low-frequency gravity component and a band-passed body-motion component using Butterworth filters. The main idea is that gravity (which encodes device orientation) and body acceleration (which encodes actual motion) carry physically different information and should be modeled as separate channels rather than as a single mixed signal. The method collects 3-axis accelerometer and gyroscope data at 50 Hz from 30 volunteers performing 6 activities, applies a 3rd-order Butterworth low-pass filter with a 0.3 Hz cutoff to extract gravity, computes the body component as raw minus gravity, and trains a multiclass SVM on features extracted from 2.56-second windows with 50% overlap. The approach is evaluated on the resulting UCI HAR benchmark using a 70/30 random split across subjects. Results show that the SVM achieves about 96% test accuracy, and that omitting the gravity/body decomposition causes features to encode device orientation as strongly as activity, leaking subject identity into the model. However, limitations include reliance on a single waist-mounted device rather than a wrist sensor (where transient orientation changes are larger), and an evaluation that does not test cross-subject generalization beyond a single random split. Overall, the paper shows that explicit gravity/body decomposition is a necessary preprocessing

step for accelerometer-based HAR and remains the canonical precedent for any pipeline using inertial sensors. [8]

The paper *Physical Human Activity Recognition Using Wearable Sensors* by Ferhat Attal, Samer Mohammed, Mariam Dedabrishvili, Faicel Chamroukhi, Lhassane Oukhellou, and Yacine Amirat provides a comprehensive review of activity recognition systems based on wearable inertial sensors. The main idea is that human activities can be effectively classified by transforming raw accelerometer signals into meaningful statistical and spectral features before applying machine learning models. The paper outlines a standard pipeline consisting of data acquisition, preprocessing, segmentation, feature extraction, and classification. In particular, the authors emphasize the importance of time-domain features such as mean and standard deviation, as well as frequency-domain features such as spectral energy and dominant frequency, which capture motion intensity and periodicity. The methods discussed include traditional classifiers such as Support Vector Machines, Decision Trees, and k-Nearest Neighbors applied to windowed sensor data. The study evaluates performance across multiple datasets and shows that carefully engineered features can significantly improve classification accuracy. However, limitations include dependence on handcrafted features and sensitivity to sensor placement and noise. Overall, the paper demonstrates that feature extraction is a critical step in converting raw motion signals into structured representations for activity recognition, which directly supports the feature engineering approach used in our project. [9]. The paper *A Survey on Human Activity Recognition using Wearable Sensors* by Oscar D. Lara and Miguel A. Labrador presents a broad overview of methodologies used in activity recognition systems based on wearable devices. The central idea is that Human Activity Recognition can be modeled as a pipeline where raw sensor data is segmented into windows and transformed into descriptive features before classification. The survey categorizes features into time-domain and frequency-domain groups and highlights their respective roles: time-domain features capture overall motion magnitude and variability, while frequency-domain features identify periodic movement patterns. The paper reviews multiple datasets and classification approaches, including probabilistic models, decision trees, and support vector machines, and evaluates their effectiveness across different applications such as health-care monitoring and smart environments. Results show that combining multiple feature types generally improves classification performance compared to using a single type alone. However, the authors note limitations such as computational cost, feature redundancy, and challenges in real-world deployment due to user variability. Overall, the paper reinforces the importance of extracting both statistical and spectral features from accelerometer data, which aligns with our projects use of mean, RMS, standard deviation, dominant frequency, spectral energy, and spectral entropy to represent human motion patterns. [10].

The paper *How Validation Methodology Influences Human Activity Recognition Mobile Systems* by Bragana, Colonna, Oliveira, and Souto studies how different validation methods affect reported Human Activity Recognition performance. The main idea is that the way a dataset is split can strongly change the meaning of the results. In HAR, windows from the same subject are often highly related because the same person may perform activities with consistent movement speed, wrist position, strength, and rhythm. If random holdout or regular k-fold validation is used, samples from the same subject can appear in both training and testing. This can make the model look more accurate than it would be on a new user. The authors compare validation strategies and show why subject-independent evaluation is important when the goal is real-world generalization. This directly supports our use of Leave-One-Subject-Out evaluation because our dataset has only three users, and our model should be tested on whether it can classify activities from an unseen participant. A limitation is that LOSO can produce noisier estimates when there are very few subjects, which also applies to our project. Still, this paper

supports the idea that LOSO is a more honest evaluation method for wearable HAR than a random split alone. [11]

The paper *Human Activity Recognition: A Comparative Study of Validation Methods and Impact of Feature Extraction in Wearable Sensors* by Rehman, Ali, Khan, and Okpala compares validation methods and feature extraction choices for wearable sensor-based HAR. The paper is useful for our project because it focuses on the same type of issue we evaluated: whether a model performs well only under a subject-dependent split or whether it can generalize to a completely unseen subject. The authors discuss Leave-One-Subject-Out cross-validation as a subject-independent method, which matches the purpose of our LOSO evaluation. This supports our decision to report both the standard 80/20 split and LOSO instead of relying only on the higher random-split accuracy. The paper also reinforces the importance of windowed data and handcrafted features, since feature extraction choices can change model performance. This connects to our window-size sweep from 0.5 seconds to 5.0 seconds and our use of time-domain and frequency-domain accelerometer features. A limitation is that results from one wearable HAR dataset may not transfer directly to another dataset because sensors, users, and activities differ. However, the paper supports our overall evaluation process because it shows that validation method, feature extraction, and subject-independent testing are important parts of building a realistic HAR model. [12]

3 Methodology

There are 2 types of data collected from participants in 7 different activities. Audio data was captured by a smartphone, Sony XPERIA at a sampling rate of 8000 Hz. Raw signals are divided into 3-second-long segments. These 3 second segments were further divided into 1 second sub-segments (small segments have been shown to produce better results)[4][3] Accelerometer data was captured by a wristband, Microsoft Band 2 and its internal accelerometer at a sampling rate of 31 Hz. The captured accelerometer data was transferred via Bluetooth to the XPERIA with timestamps. The activities were Watch TV, Eat Chips, Wash Hands, Brush Teeth, Type on Keyboard, Sweep, and Mop Floor.

4 Dataset Exploration

The dataset itself contains audio and accelerometer data. Audio data features are labelled v1 and accelerometer data features are labelled v2. The raw accelerometer data was available by user but only the audio features were available due to privacy considerations to avoid sharing sensitive and personally identifiable information.

Table 1: V1 Audio MFCC Features - head of data

userid	label	v1_mfcc1	v1_mfcc2	v1_mfcc3	v1_mfcc4	v1_mfcc5	v1_mfcc6	v1_mfcc7	v1_mfcc8
user1	watch_tv	120.0789	3.0252	0.3905	-1.7778	-9.2161	-6.5676	-9.9188	-9.1455
user1	watch_tv	130.2431	2.5523	-6.4579	-7.4402	-3.0701	-3.4755	-5.7188	-2.8782
user1	watch_tv	122.2592	1.9703	-8.9806	-10.2220	-3.5650	-2.4969	-5.5849	-6.7306
user1	watch_tv	122.6096	3.1887	-7.0494	-10.5699	-0.2208	-0.6495	-4.9758	-6.8563
user1	watch_tv	121.3930	1.3388	-8.7062	-8.6550	-4.9353	-3.4017	-5.0314	-7.9579

Table 2: V2 Accelerometer Magnitude and Correlation Features - head of data

userid	label	v2_maxX	v2_maxY	v2_maxZ	v2_corXY	v2_corXZ	v2_corYZ	v2_meanMagnitude	v2_sdMagnitude	v2_auc	v2_meanDif
user1	watch_tv	0.0974	1.0881	0.2271	0.9002	-0.7295	-0.8635	1.0148	0.0604	70.0215	-0.0004
user1	watch_tv	0.7036	1.4248	0.8181	-0.2104	0.3196	-0.7225	1.0057	0.1626	79.4513	-0.0062
user1	watch_tv	0.3125	0.9727	-0.0703	-0.0189	-0.0939	-0.3057	1.0028	0.0061	77.2163	0.0000
user1	watch_tv	0.3499	1.0200	-0.0532	0.4924	0.2731	-0.0231	1.0029	0.0213	75.2159	-0.0000
user1	watch_tv	0.3223	0.9551	-0.0769	-0.2263	-0.2174	-0.0539	1.0034	0.0028	71.2433	-0.0001

Table 3: **Dataset Summary Table.** Summary table containing dataset demographics. There were 1386 total audio instances of 3 seconds each for a total of 4158 seconds of data, divide by 3 Participants averaging 1386 seconds per participant which is 0.01604167 days per participant. Users 1 and 2 had 466 records and user3 had 454. The paper [4] reported around 3 minutes of data for each of the 7 activities was collected for an estimate of 21 minutes per participant. In actuality, there was on average 23.1 minutes of data per participant.

Summary Table	
Number of Participants	3
Days Per Participant	0.00208333
Age Range	25-30
Male Participants	2
Female Participants	1
Total Labels	1386

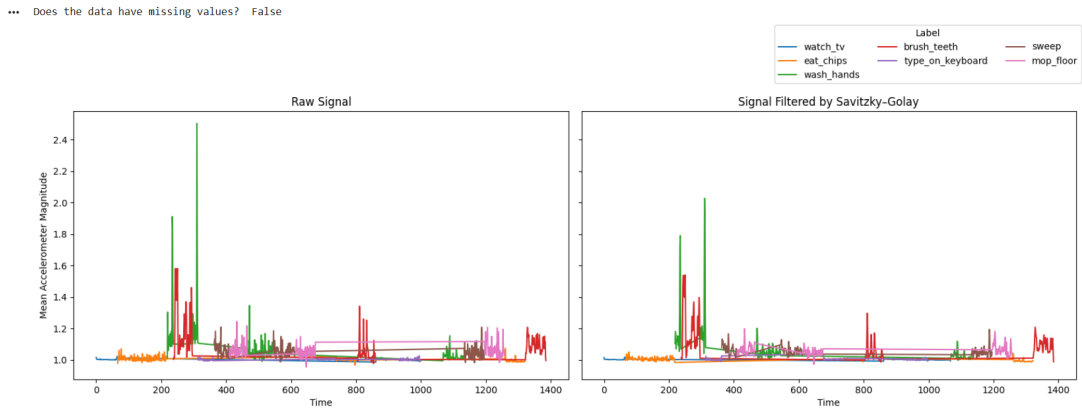


Figure 1: **Raw Accelerometer Mean Magnitude and Savitzky-Golay.**

Time-domain plots showed clear differences among activities’ acceleration. Watch TV was mostly stable with small fluctuations, indicating low movement. Eat chips showed frequent repeated fluctuations, suggesting moderate periodic hand motion. Wash hands exhibited the largest and most irregular variations, indicating vigorous movement. Other activities such as brush teeth, sweep, and mop floor also showed larger variations than passive tasks. These visual differences suggest that feature extraction over body acceleration should be able to separate low-motion from high-motion activities.

5 Annotation and Signal Exploration

Each accelerometer recording carries one of seven activity labels (`watch_tv`, `eat_chips`, `wash_hands`, `brush_teeth`, `type_on_keyboard`, `sweep`, `mop_floor`) and one of three subject IDs, with every sample inheriting the recording-level class [4]. Class counts are balanced per subject (60–70 instances per activity), supporting stratified and leave-one-subject-out evaluation without re-weighting. Our pipeline operates on the raw accelerometer streams; the V1 audio MFCC and V2 summary features in Section 4 are descriptive only. Raw traces contain occasional dropped samples and isolated single-sample glitches, both repaired in cleanup (Section 6) via linear interpolation and a length-3 median filter.

Grouping the per-segment summary `v2_meanMagnitude` by label and overlaying a Savitzky-Golay smoother (window 5, order 2) reveals the expected passive/active split: `watch_tv` and `type_on_keyboard` cluster around the ~ 1 g gravity baseline, whereas `sweep`, `mop_floor`, `wash_hands`, and `brush_teeth` show substantially larger spread. Even a single scalar per segment carries class-discriminative information, motivating the sample-level gravity/body

decomposition in Section 6.

6 Signal Preprocessing

Raw accelerometer recordings for the seven home-task activities and the three subjects (`user1`, `user2`, `user3`) were read from per-user `.txt` files containing four columns (*timestamp*, x , y , z). Timestamps were converted to relative seconds, and the scalar magnitude $m = \sqrt{x^2 + y^2 + z^2}$ was computed per sample. The overall pipeline (preprocessing $\rightarrow$ segmentation $\rightarrow$ feature extraction $\rightarrow$ classification) follows the Activity Recognition Chain [13], with the per-axis preprocessing adapted from the gravity/body decomposition standard in HAR literature [14].

Cleanup and resampling. Each trace was sorted by timestamp and missing samples in $\{x, y, z, m\}$ were repaired with linear interpolation followed by backward/forward fill. Because the Microsoft Band 2 does not deliver strictly uniform samples (nominal $f_s = 31$ Hz with variable inter-sample intervals), each trace was then linearly resampled onto a fixed 30 Hz grid so that window lengths and DFT bin spacings are identical across recordings and subjects. Single-sample sensor glitches were then removed with a length-3 median filter applied on the uniform grid,

$$\tilde{a}[n] = \text{median}(a[n-1], a[n], a[n+1]), \quad (1)$$

chosen over a mean kernel for its robustness to outliers: a mean kernel would smear each spike into its neighbours, whereas the median in Equation 1 rejects them outright. The effective sampling rate is therefore $f_s = 30$ Hz throughout.

Gravity / body decomposition. Wrist accelerometer signals mix two physically distinct components: gravity (slow, near-DC, encoding wrist orientation) and body acceleration (the actual motion of interest) [8]. To separate them, each axis $a \in \{x, y, z\}$ was passed through a 4th-order zero-phase Butterworth low-pass filter with cutoff $f_g = 0.5$ Hz to obtain the gravity component a_g (matching the UCI HAR convention [8]). The body component

$$a_b = a - a_g \quad (2)$$

was then band-passed with a 4th-order Butterworth filter at $[0.25, 12]$ Hz, the band of typical human-motion frequencies. The body magnitude

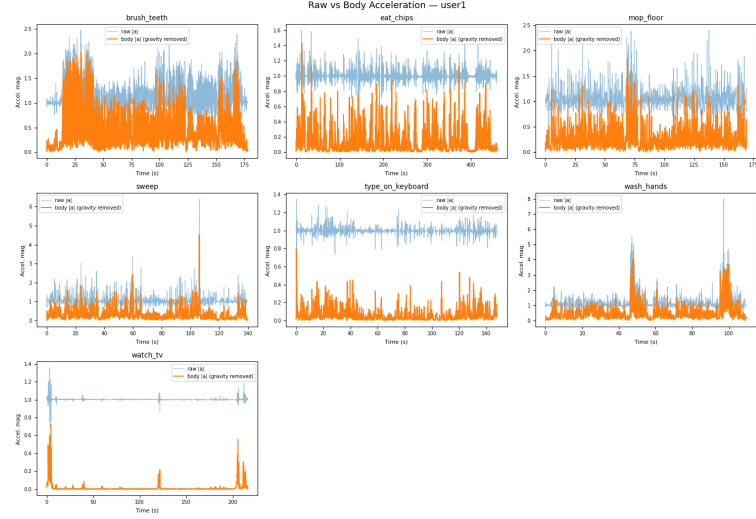
$$m_b = \sqrt{x_b^2 + y_b^2 + z_b^2} \quad (3)$$

sits near zero at rest and produces clear motion bursts during activity, in contrast to m which is dominated by the ~ 1 g gravity baseline regardless of activity.

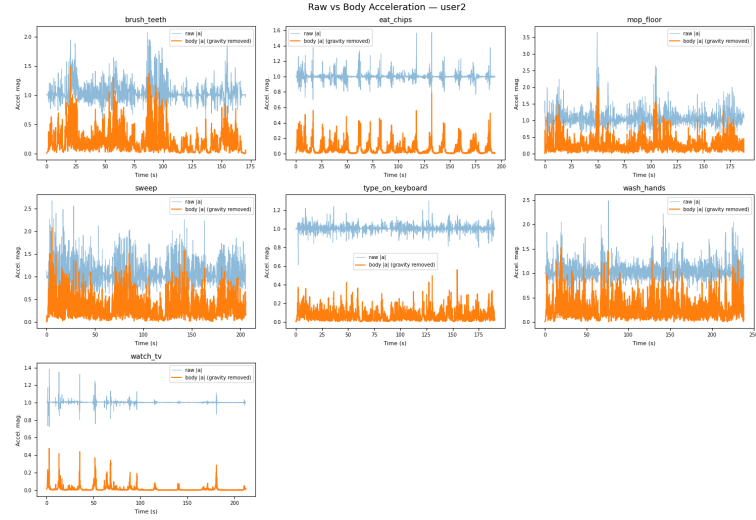
The body magnitude m_b is the single direction-invariant scalar quantifying instantaneous wrist motion: per-axis amplitudes vary with stroke direction whereas m_b does not, removing a nuisance source of variance. The gravity magnitude $m_g = \sqrt{x_g^2 + y_g^2 + z_g^2}$ remains ≈ 1 g across activities and is used only as a sanity check that the decomposition has been applied correctly. All time- and frequency-domain features (Section 8) are computed on m_b .

Removing gravity is critical for cross-subject generalization: each subject wears the wristband at a slightly different orientation, so gravity-axis features encode subject identity rather than activity. Empirically, including gravity-axis features inflated random-split accuracy but collapsed leave-one-subject-out accuracy (Section 10). Only the body channels $\{x_b, y_b, z_b, m_b\}$ are passed to the feature pipeline.

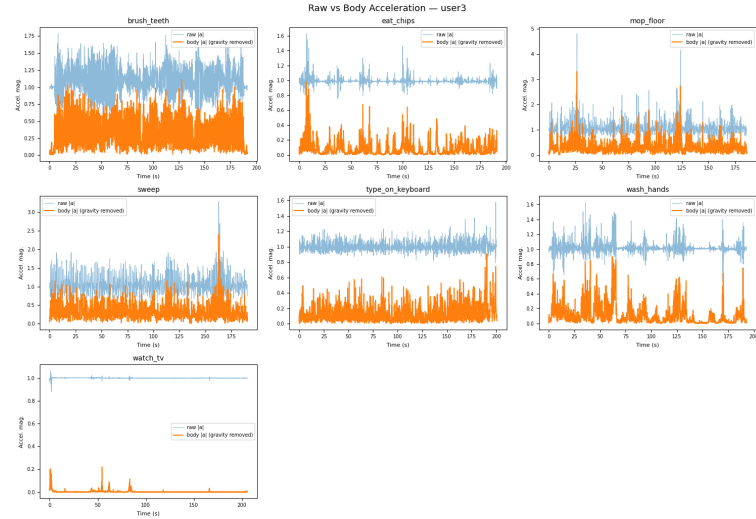
Figure 2 contrasts the raw magnitude m and the body magnitude m_b for the seven activities, per subject. The raw line hovers around the gravity baseline and varies only subtly with motion; the body line sits near zero at rest and produces activity-specific bursts whose amplitude separates passive (`watch_tv`, `type_on_keyboard`) from active (`brush_teeth`, `sweep`, `wash_hands`, `mop_floor`) tasks. Between-user variation in burst amplitude and timing is also visible.



(a) user1



(b) user2



(c) user3

Figure 2: **Raw vs. body acceleration magnitude per subject.** Light line: raw magnitude m ; dark line: gravity-removed body magnitude m_b (Equation 3). The raw line hovers around the ~ 1 g gravity baseline; the body line sits near zero at rest and bursts during motion, separating passive from active tasks.

7 Windowing Strategies

Window length W controls a stability–temporal-resolution trade-off [7]: short windows are noisy but yield many training samples, long windows are stable but lose granularity and sample count. We treated W as a hyper-parameter and selected it empirically.

Protocol. For each user the body magnitude m_b (Equation 3) of every activity was segmented with a sliding window of $W f_s$ samples and 50% overlap. Ten candidate window lengths were tested, $W \in \{0.5, 1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0, 4.5, 5.0\}$ s. For the k -th window of activity a we computed a normalised energy

$$E_{a,k}(W) = \frac{1}{W f_s} \sum_{i=1}^{W f_s} m_{b,i}^2, \quad (4)$$

and used the coefficient of variation $CV_a(W) = \sigma(E_{a,\cdot})/\mu(E_{a,\cdot})$ across windows of the same activity as a stability measure: low CV means successive windows produce similar energy values (stable segmentation), high CV means within-activity variability still dominates. For each user we report the mean of $CV_a(W)$ over the seven activities, together with the mean number of windows per recording, in Figure 3.

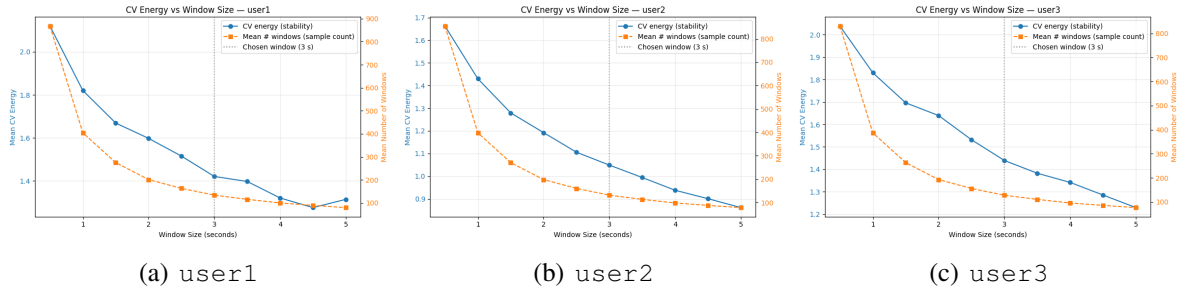


Figure 3: **Window-size selection per subject.** Left axis (solid): mean coefficient of variation of the per-window normalised energy across activities (Equation 4). Right axis (dashed): mean number of windows generated per recording. The vertical dotted line marks the chosen window size ($W = 3$ s). All three users exhibit the same monotonic decrease in CV with window length, with the steepest drop before $W = 2$ s, after which the curve flattens substantially while the number of windows continues to drop. User2 has the lowest overall CV (cleaner signal across activities); user1 and user3 are at a comparable, slightly higher baseline.

Per-user observations. Mean CV falls from 2.12 at $W = 0.5$ s to 1.31 at $W = 5$ s for user1; from 1.66 to 0.86 for user2; from 2.04 to 1.23 for user3. The steepest drop occurs before $W = 2$ s for all three users, after which the curves flatten substantially with only marginal further improvement. User2 has the lowest overall CV; user1 and user3 are comparable. The mean number of windows per recording drops from several hundred at $W = 0.5$ s to fewer than 100 at $W = 5$ s, confirming the stability–sample-count trade-off.

Choice of window size. The CV-knee sits consistently between 1 and 2 s for every subject, so any $W \in [2, 3]$ s lies on the flat low-CV portion. We adopted $W = 3$ s with 50% overlap: past the knee for stable energy statistics, sufficient samples for reliable spectral feature estimation (dominant frequency, spectral entropy), long enough to capture a full cycle of periodic activities (mop, sweep, brush), and yielding enough windows per recording for model training. This choice is empirically validated by the model-based window sweep in Section 10.

8 Feature Extraction

Raw accelerometer windows obtained from the preprocessing and segmentation pipeline were transformed into compact numerical descriptors for machine learning. Feature extraction is an important step in Human Activity Recognition (HAR) because raw time-series signals are high-dimensional and often noisy. By summarizing each window into informative statistics, models can more effectively distinguish between household activities such as sweeping, brushing teeth, and watching TV [9, 10].

Time-domain features. For each segmented window, four time-domain features were extracted from the body acceleration magnitude signal m_b . The mean measures the average motion intensity over the window, while the standard deviation measures signal variability. Root Mean Square (RMS) captures signal power and overall movement energy. Zero Crossing Rate (ZCR) counts how frequently the centered signal changes sign and reflects motion fluctuation or rhythm. These features are computationally efficient and commonly used in wearable sensor classification tasks [9].

Frequency-domain features. To capture periodic movement patterns, each window was transformed into the frequency domain using the Fast Fourier Transform (FFT). Three frequency-domain features were extracted: dominant frequency, spectral energy, and spectral entropy. Dominant frequency identifies the strongest repeated motion cycle, which is useful for repetitive activities such as sweeping or mopping. Spectral energy measures total oscillation strength, while spectral entropy quantifies the complexity or irregularity of the movement spectrum. Frequency-domain features often improve recognition performance when activities contain repeated motion patterns [10].

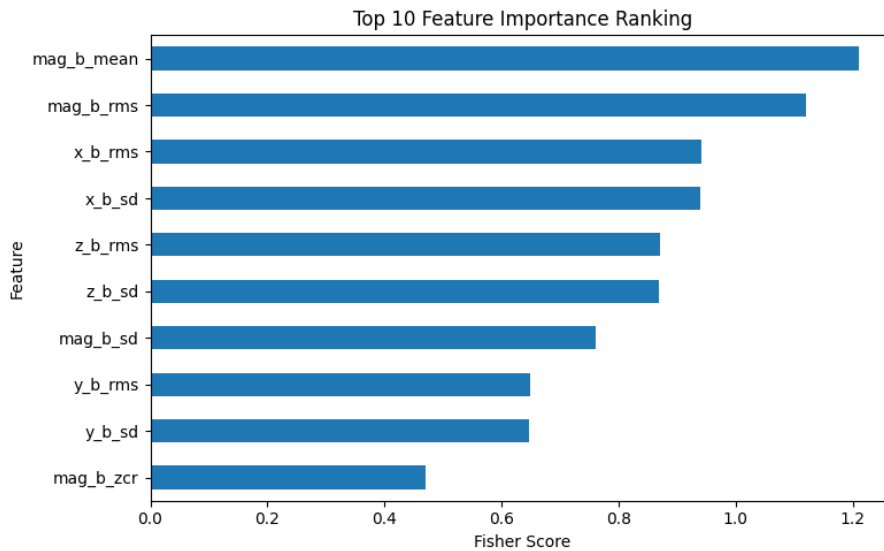


Figure 4: **Feature importance ranking using Fisher score.** Higher-ranked features contributed more strongly to distinguishing household activities. Energy-based and variability-based features were especially useful for separating passive activities from active or repetitive movements.

Interpretation. The extracted features reveal clear differences between passive and active behaviors. Low-motion activities such as watching TV and typing on a keyboard generally produced lower RMS and standard deviation values, while more vigorous tasks such as brushing

teeth, sweeping, washing hands, and mopping produced larger values. Frequency-domain features further helped distinguish repetitive arm motions from irregular actions. Combining time-domain and frequency-domain features therefore provides a richer representation than either feature type alone.

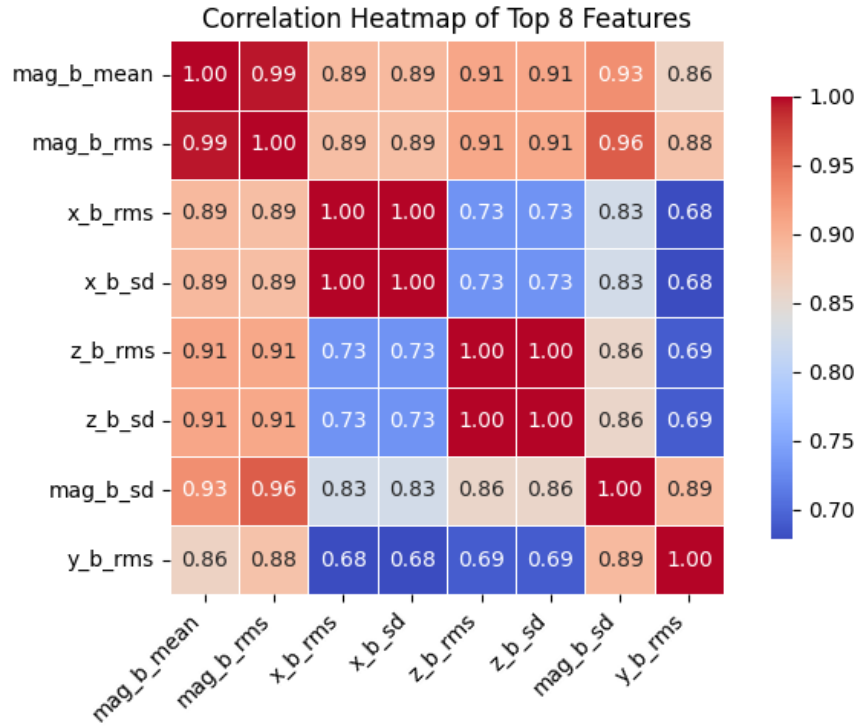


Figure 5: **Correlation heatmap of extracted features.** The heatmap shows relationships among the extracted time-domain and frequency-domain features. Movement intensity features such as RMS, standard deviation, and spectral energy are expected to be related, while frequency-based features provide additional information about motion rhythm and complexity. We selected a subset of features based on their importance using Fisher score ranking. The heatmap therefore focuses on the most informative features, allowing clearer interpretation of relationships without losing key insights.

Feature importance analysis. A Fisher score ranking method was applied to estimate the discriminative value of each extracted feature. For readability, Figure 4 displays only the top-ranked features, despite in the notebook, we made a total of 31 features. This was based off from the fisher score ranking, and it was limitedly chosen based on the ranking, so we can see the graph easier. Higher-scoring features contribute more strongly to separating activity classes. In our experiments, energy-based and variability-based measures such as RMS, standard deviation, and spectral energy ranked highly, indicating that movement intensity is a strong indicator of activity type.

The feature engineering pipeline reduced raw sensor windows into meaningful descriptors that improved interpretability and provided effective inputs for the machine learning models evaluated in later sections.

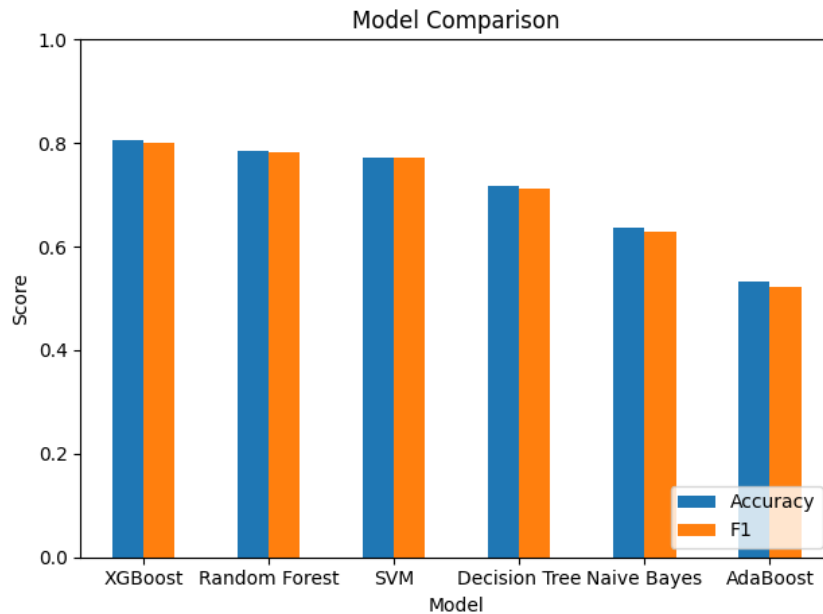
9 Modeling and Results

Model Descriptions. We implemented classical ML models: Decision Tree, SVM, Naive Bayes, Random Forest, AdaBoost, and XGBoost. Decision trees help make decisions by mapping out different choices and their possible outcomes. There are two types of decision trees: classification and regression. SVM is a supervised machine learning algorithm used for

regression and classification tasks. The goal of SVM is to find the margin between classes. Naive Bayes is an algorithm that predicts the category of a data point based on probabilities. Random Forest is an algorithm that uses many decision trees to improve predictive performance. AdaBoost is a boosting technique that combines several weak classifiers in sequence to build a strong one. Each new model focuses on correcting the mistakes of the previous one until all data is correctly classified or a set number of iterations is reached. XGBoost is an optimized gradient boosting algorithm that combines multiple weak models into a stronger, high-performance model.

Model	Accuracy	Precision	Recall	F1
XGBoost	0.8036	0.8015	0.7972	0.7983
Random Forest	0.7856	0.8041	0.7751	0.7837
SVM	0.7730	0.7807	0.7712	0.7728
Decision Tree	0.7171	0.7130	0.7143	0.7121
Naive Bayes	0.6360	0.6633	0.6409	0.6298
AdaBoost	0.5333	0.5519	0.5348	0.5224

Table 4: **Model comparison under standard 80/20 stratified random split.** Six classical ML models trained on the full 31-feature set (body acceleration + cross-axis correlations) with class-balanced sample weights. Standard split mixes windows from all three subjects across train and test; the realistic cross-subject (LOSO) numbers are reported in Section 10, Table 5.



(a) Model Comparison

Model Results. We trained all models on an 80/20 split and standardized all features using the standard scaler. Among the 6 models, XGBoost achieved the best results, with an accuracy of 0.8036 and an F1 score of 0.7983. XGBoost’s strength lies in its ability to capture complex, non-linear relationships between features, making it the most reliable classifier for identifying the behavior occurring at any given moment. This model converged faster and was less prone to overfitting than the other models. [5]

Deep learning preprocessing. For sequence ANN architectures (1D-CNN, LSTM, GRU) we re-emit the same per-window data without the feature-aggregation step, producing the tensor $X_{dl} \in \mathbb{R}^{2774 \times 90 \times 4}$ in (batch, time, channels) order over the four body channels $\{x_b, y_b, z_b, m_b\}$, paired with $y_{dl} \in \mathbb{R}^{2774}$ [6]. The window count matches the classical row count at $W = 3$ s, so the two pipelines share a partition. Figure 6 verifies window–label alignment and channel non-redundancy on one representative window per activity. Training a sequence model on X_{dl} is left as future work (Section 11).

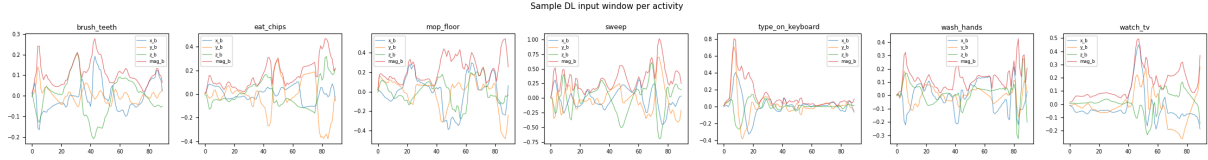


Figure 6: **Deep-learning preprocessing sanity check.** One representative 3-second window per activity drawn from the $(2774, 90, 4)$ tensor X_{dl} . Each panel overlays the four body-acceleration channels $\{x_b, y_b, z_b, m_b\}$ over the $W f_s = 90$ samples of one window. Passive activities (watch_tv, type_on_keyboard) sit near zero; moderate activities (brush_teeth, eat_chips) show repetitive structure; vigorous activities (sweep, mop_floor, wash_hands) show large-amplitude periodic excursions on multiple axes. The plot verifies window–label alignment, propagation of gravity removal into the deep-learning tensor, and that the four channels carry non-redundant information.

10 Results

Headline metrics. Table 5 reports tuned XGBoost performance under both evaluation protocols. Under the standard 80/20 stratified split, the model achieved 81.8% accuracy and 81.3% macro F1. Under Leave-One-Subject-Out (LOSO) evaluation across the three subjects, the model achieved $62.5\% \pm 1.0\%$ accuracy and $60.9\% \pm 0.9\%$ macro F1. This shows that the model performs better when windows from the same users can appear in both training and testing, while LOSO provides a stricter estimate of cross-user generalization.

Table 5: **Final tuned XGBoost metrics under both evaluation protocols.** The standard 80/20 stratified split mixes windows from all three subjects across train and test; LOSO holds one entire subject out per fold and averages across the three folds.

Metric	Standard 80/20	LOSO mean	LOSO std
Accuracy	0.8180	0.6251	0.0103
Precision (macro)	0.8123	0.6276	0.0115
Recall (macro)	0.8143	0.6314	0.0293
F1 (macro)	0.8128	0.6090	0.0091

Window-size sweep. The same XGBoost classifier was trained at ten window lengths from 0.5 to 5.0 s under both protocols (Table 6). Both curves rise steeply between 1.5 and 2.5 s and begin to plateau afterward. The best standard split accuracy occurred at $W = 4.5$ s, while the best LOSO accuracy occurred at $W = 5.0$ s. However, the chosen $W = 3$ s remains a reasonable choice because it performs close to the stronger window sizes while retaining 2774 windows compared to only 1661 windows at $W = 5$ s. This supports the idea that 3 seconds balances motion context, model performance, and available training samples.

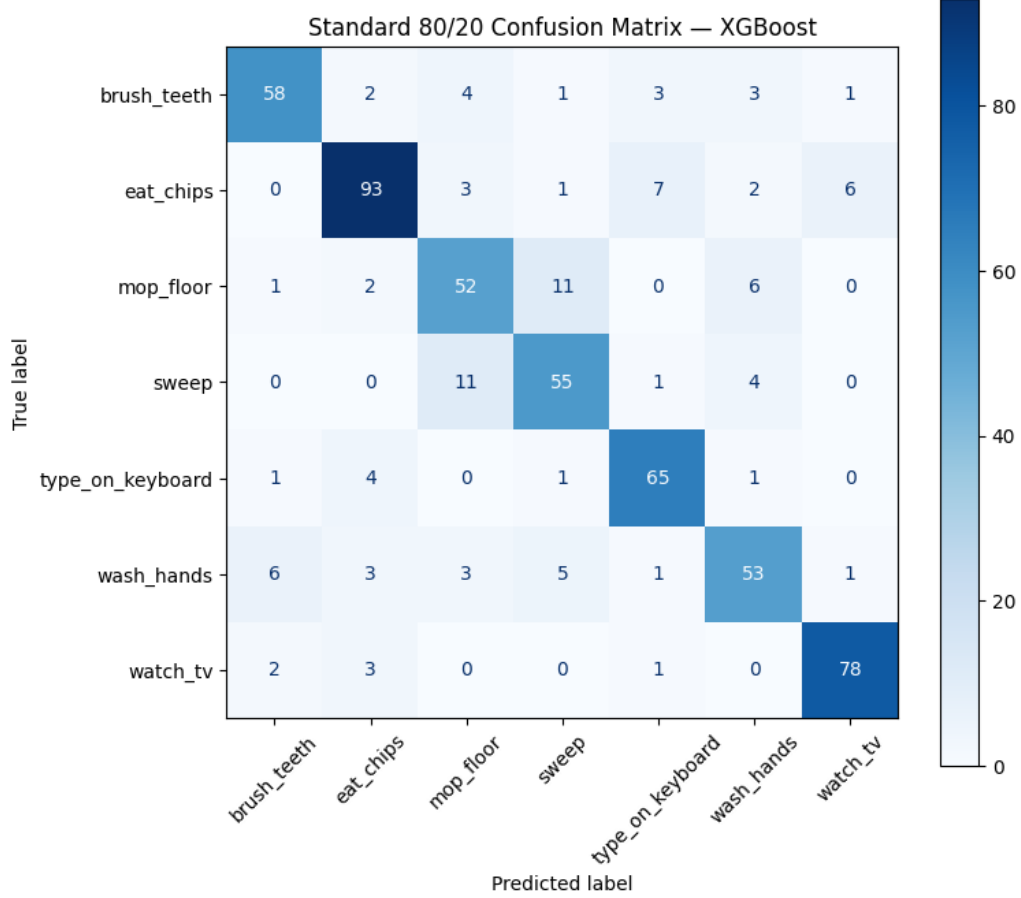


Figure 7: Standard 80/20 split confusion matrix for the tuned XGBoost model. This evaluation gives higher performance because windows from the same users can appear in both training and testing.

Table 6: **Window-size sweep.** Ten window sizes evaluated under both the standard 80/20 split and LOSO. The chosen $W = 3$ s row is highlighted. Accuracy generally increases with window length but begins to plateau after around 3 s.

W (s)	n_windows	Split acc	LOSO acc (mean $\pm$ std)	LOSO F1
0.5	17895	0.665	0.505 $\pm$ 0.039	0.493
1.0	8345	0.721	0.549 $\pm$ 0.034	0.539
1.5	5686	0.765	0.581 $\pm$ 0.035	0.565
2.0	4167	0.783	0.613 $\pm$ 0.033	0.597
2.5	3377	0.790	0.605 $\pm$ 0.027	0.592
3.0	2774	0.805	0.609 $\pm$ 0.032	0.593
3.5	2400	0.792	0.629 $\pm$ 0.032	0.615
4.0	2078	0.784	0.631 $\pm$ 0.036	0.621
4.5	1859	0.841	0.632 $\pm$ 0.032	0.618
5.0	1661	0.835	0.643 $\pm$ 0.028	0.630

Failure modes. The aggregated LOSO confusion matrix (Figure 10) and the error-pair summary (Table 7) show two dominant off-diagonal regions. `sweep` $\leftrightarrow$ `mop_floor` accounts for many of the most common errors because both activities involve similar rhythmic, large-amplitude arm motion. `eat_chips` is also confused with multiple classes such as `type_on_keyboard`,

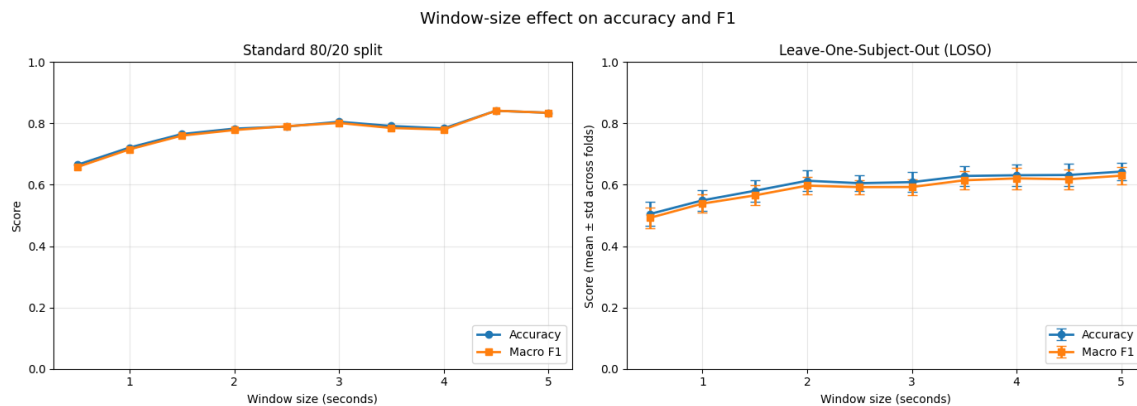


Figure 8: Window-size effect on accuracy and macro F1 across ten tested window sizes. The left plot shows the standard 80/20 split, while the right plot shows Leave-One-Subject-Out evaluation.

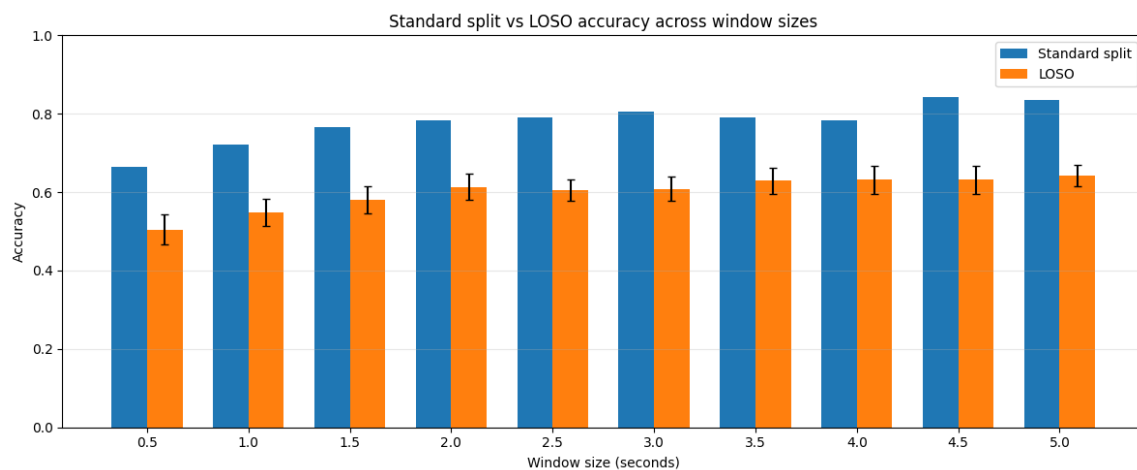


Figure 9: Standard split accuracy compared with LOSO accuracy across the ten tested window sizes. The consistent gap between the two protocols shows that random splitting gives a higher and less realistic estimate than subject-independent testing.

`watch_tv`, and `wash_hands`. This makes sense because a 3 s window can capture either a rest phase, which looks passive, or a single hand-to-mouth reach, which looks more active.

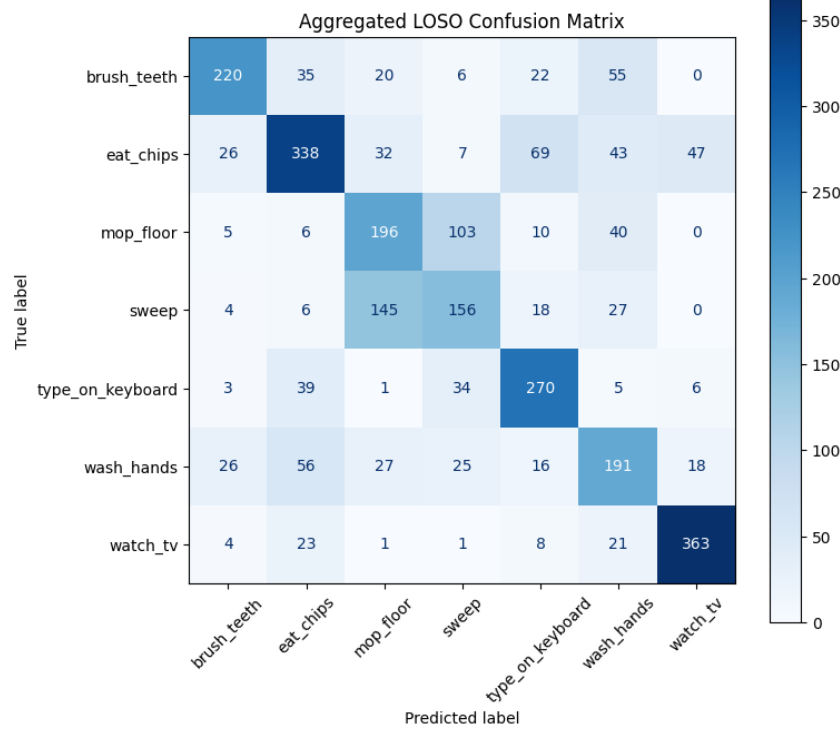


Figure 10: **Aggregated LOSO confusion matrix** (XGBoost, predictions pooled across all three folds). Rows are true activity labels, columns are predicted labels; brighter cells indicate larger counts. The strong off-diagonal blocks at sweep / mop_floor and around eat_chips correspond to the failure modes discussed in the text.

Table 7: **Most common LOSO error pairs from the aggregated confusion matrix.** The table lists the most frequent cases where the model predicted the wrong activity during subject-independent testing.

True Label	Predicted Label	Count
sweep	mop_floor	145
mop_floor	sweep	103
eat_chips	type_on_keyboard	69
wash_hands	eat_chips	56
brush_teeth	wash_hands	55
eat_chips	watch_tv	47
eat_chips	wash_hands	43
mop_floor	wash_hands	40
type_on_keyboard	eat_chips	39
brush_teeth	eat_chips	35

11 Discussion

Standard split overstates deployment performance. The gap between the two reported numbers (81.8% vs 62.5%, same tuned XGBoost model) is the central methodological finding. The standard random split mixes windows from all subjects across train and test, allowing the model to partly classify by subject identity, such as typical motion amplitudes, wrist-mounting orientation, and characteristic timing, rather than activity alone. Any deployment scenario, such as a wearable classifying activities for a new user, corresponds more closely to the LOSO setting than the random split. Reporting only the random-split number would substantially overstate real-world capability.

Cross-subject consistency. The LOSO standard deviation is useful because it shows how much performance changes across held-out users. In our final evaluation, the LOSO accuracy standard deviation was 0.0103, which means the model’s performance was fairly consistent across the three subjects. This suggests the model was not relying on only one easy subject. The gravity/body decomposition (Section 6) likely contributes by removing the wrist-orientation component that varies most across subjects.

Failure modes have physical explanations. `sweep` $\leftrightarrow$ `mop_floor` would require either much richer features such as time-series shape descriptors and motif analysis, or auxiliary signals such as audio of the broom versus the mop, to resolve more accurately. `eat_chips` is bimodal at the 3 s timescale and would be better handled by sequence models such as an LSTM or CNN operating on raw windowed signals rather than only per-window summary statistics.

Limitations. Three caveats apply. First, only three subjects were available, so LOSO has just three folds and per-fold variance is intrinsically noisier than larger HAR benchmarks. Second, only wrist accelerometer data was used in the final evaluation, while the dataset also includes audio features that were not fully exploited. Third, the feature set is hand-crafted; classical ML approaches can be interpretable and efficient, but end-to-end deep models may capture more complex temporal patterns. The deep-learning preprocessing pipeline provides a $(n_windows, window_length, n_channels)$ tensor in the format an LSTM or 1D-CNN would consume, leaving deep-model evaluation as a natural follow-up.

12 Conclusion

This project evaluated human activity recognition using wrist accelerometer signals from household activities. The full pipeline included preprocessing, sliding-window segmentation, feature extraction, classical machine learning models, and advanced evaluation. Tuned XGBoost produced the strongest standard-split result, reaching about 81.8% accuracy and 81.3% macro F1. However, LOSO mean evaluation produced lower results, with about 62.5% accuracy and 60.9% macro F1. This difference shows that the standard split is useful as a baseline, but it can overestimate real-world performance because windows from the same users appear in both training and testing.

The window-size analysis showed that very short windows were less stable, while longer windows generally improved classification performance. Results began to plateau after around 3 seconds, making the selected 3-second window a reasonable trade-off between accuracy, stability, and number of training samples. The confusion matrix and error analysis showed that many mistakes were physically understandable. Activities such as sweeping and mopping were often confused because they share repeated arm movements, while eating chips was harder to classify because a short window may capture either a rest period or a quick hand-to-mouth motion.

Overall, the results show that accelerometer-based HAR can recognize several home activities, but cross-user generalization remains challenging with only three participants and wrist accelerometer data alone. Future work could improve performance by adding audio features, increasing the number of participants, testing deep sequence models such as CNNs or LSTMs, and using longer temporal context to better capture activities with repeated or irregular motion.

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